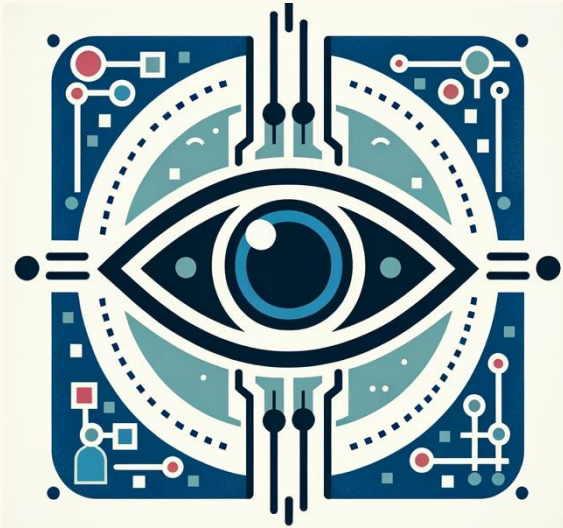


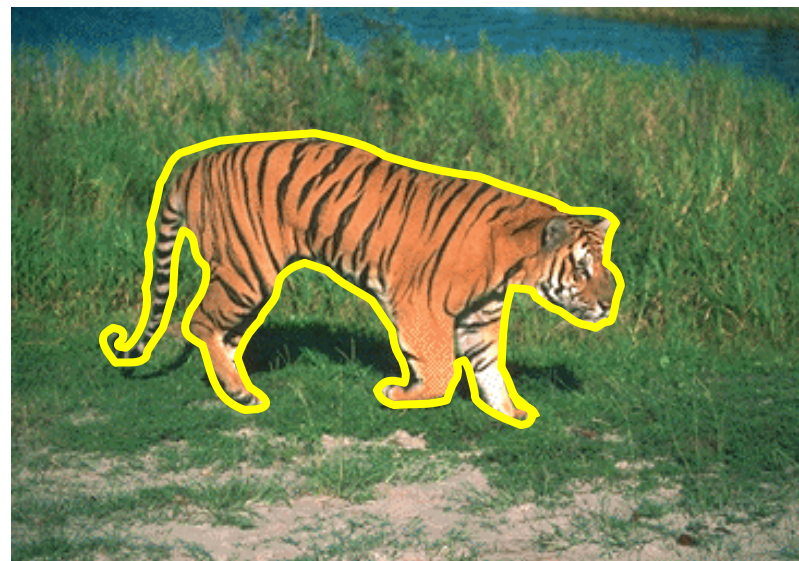
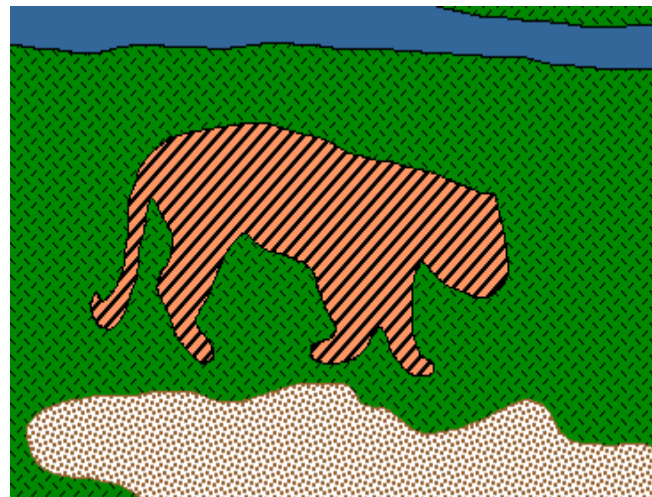


COMPUTER VISION

Segmentation

Segmentation

- Segmentation → separate **image into regions**
- A **region** is:
 - a group of connected pixels with similar properties
- But, what defines these regions?



Semantic Segmentation

Segmentation

- Segmentation → separate **image into regions**
- But, what defines these regions?
 - **Semantics:** requires high-level knowledge
 - **Appearance:** group/cluster together regions with similar characteristics (colour, texture, etc)

Segmentation



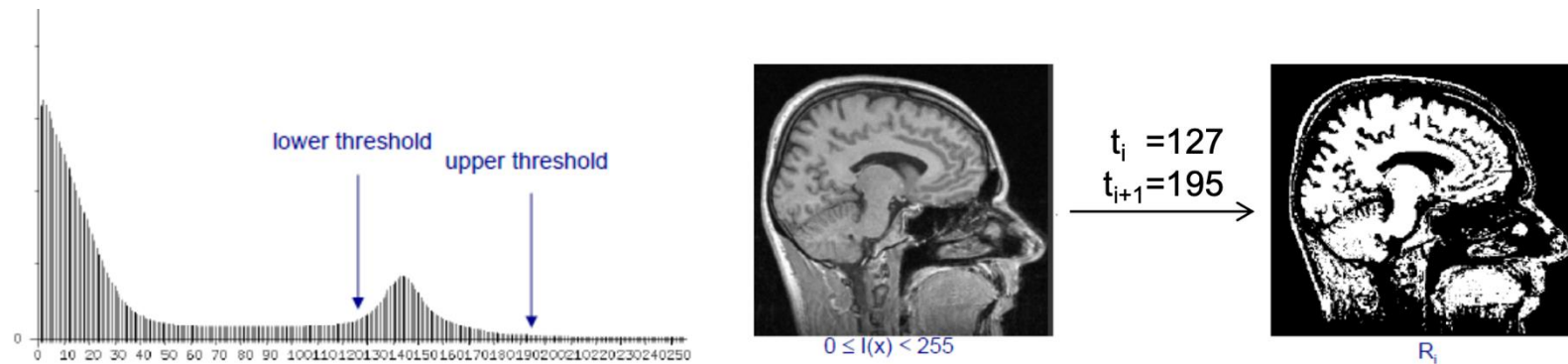
Segmentation

- **Edge-based** approaches:
 - Use the boundaries of regions to segment the image
 - (as seen before) Detect abrupt changes in intensity (discontinuities)
 - Gaps in edges have to be bridged
- **Region-based** approaches:
 - Use similarity and spatial proximity among pixels to find different regions
- Edges are found based on **differences** between values of adjacent pixels.
- Regions are found based on **similarities** between values of adjacent pixels.
- Theoretically, both approaches should give identical results but this is not true in practice

Thresholding

- Regions with uniform intensity give rise to strong peaks in the histogram
- Select **one or more** intensity values (**thresholds**)
- Partition the image domain according to the intensity being higher or lower than those thresholds

$$R_i = \{x \in D_I: t_i \leq I(x) < t_{i+1}\}$$

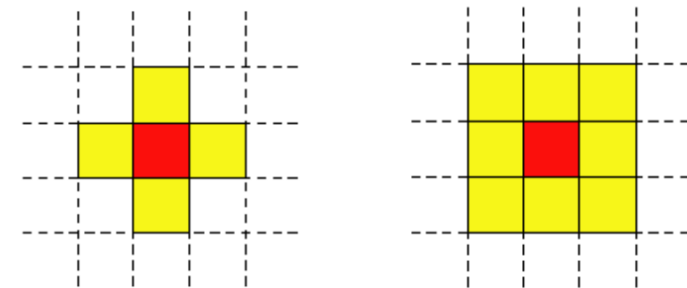


Thresholding

- Selection of thresholds:
 - Manual
 - Automatic
 - at local minima (requires smoothing of histogram first)
 - at intersection of fitted Gaussian functions
 - from prior knowledge
 - intensity characteristics of the objects
 - sizes of the objects
 - fractions of an image occupied by the objects
 - number of different types of objects appearing in an image

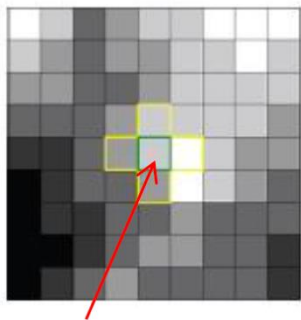
Region Growing

- Region-growing approaches exploit the important fact that pixels which are close together have similar grey values.
- General algorithm:
 - Start with a single pixel (seed) and add new pixels iteratively
 1. Choose the seed pixel
 2. Check the neighbouring pixels and add them to the region if they are similar to the seed
 3. Repeat step 2 for each of the newly added pixels; stop if no more pixels can be added.
- Crucial definitions:
 - Neighbourhood
 - Homogeneity (aggregation criterion)
 - Stopping criterion

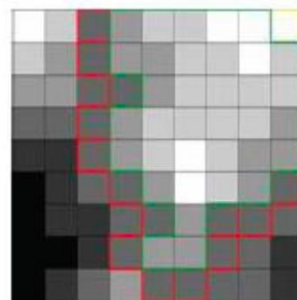
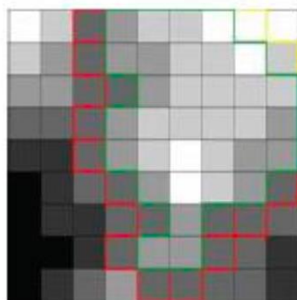
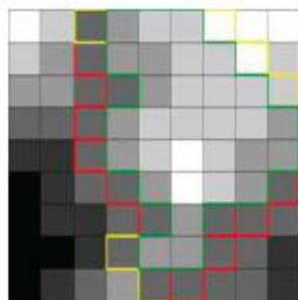
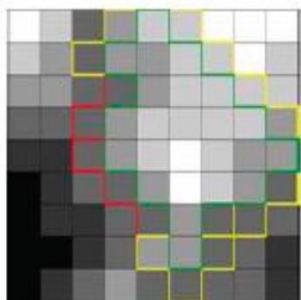
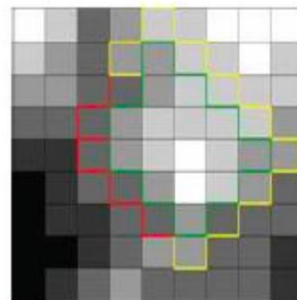
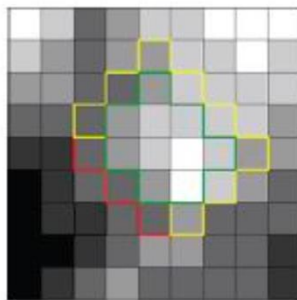
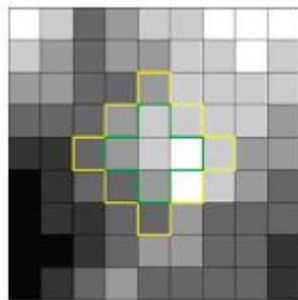


Neighborhood

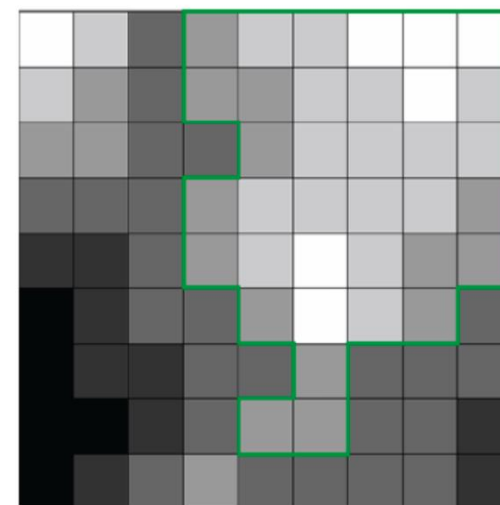
Region Growing



seed point



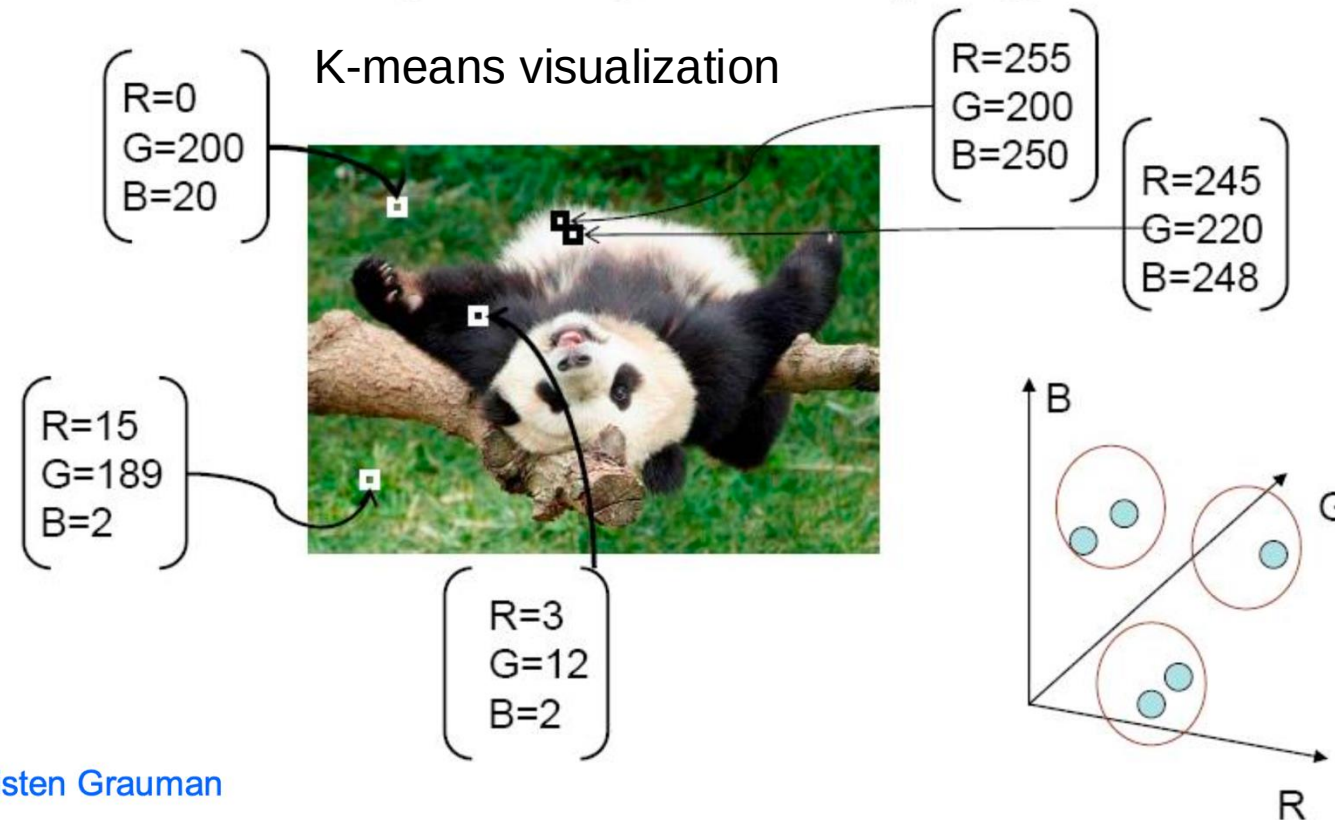
Similarity criterion: gray value difference to seed point ≤ 1 ,
neighborhood = 4



Segmentation result

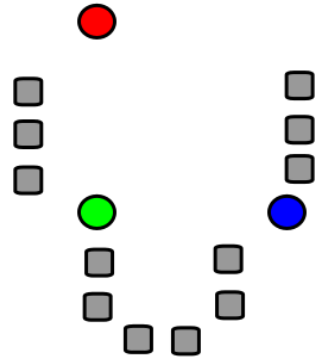
Clustering

- Cluster similar pixels (features) together

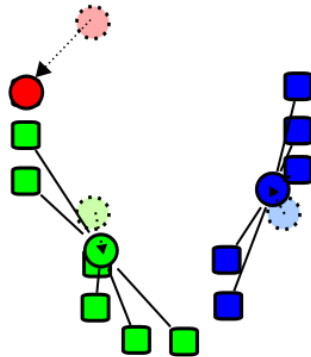


Slide credit: Kristen Grauman

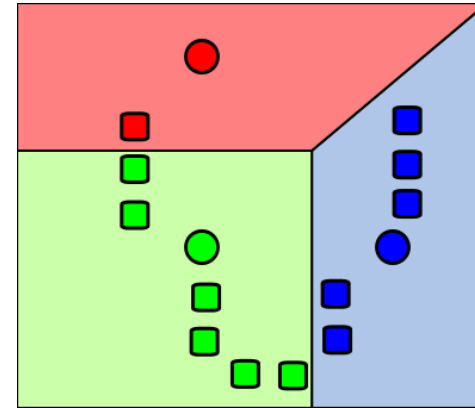
Clustering using K-means



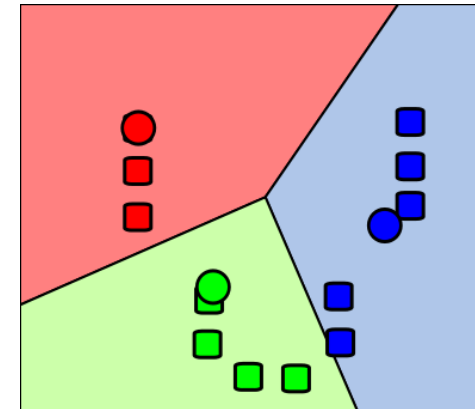
1. Select initial centroids at random



3. Compute each centroid as the mean of the objects assigned to it (go to 2)



2. Assign each object to the cluster with the nearest centroid.



2. Assign each object to the cluster with the nearest centroid.

Repeat previous 2 steps until no change

Clustering using K-means

Given k :

1. Select initial centroids at random.
2. Assign each object to the cluster with the nearest centroid.
3. Compute each centroid as the mean of the objects assigned to it.
4. Repeat previous 2 steps until no change.

How to find K ?

- Use prior knowledge about image
- Use different K 's and test for goodness of clusters
- Analyse image histograms

K-means

- K-means clustering based on intensity or colour
 - Clusters don't have to be spatially coherent
 - Clustering based on colour components & position enforces more spatial coherence



Image



Intensity-based clusters



Color-based clusters

Post-processing

- Segmentation results can be noisy
 - post-processing might be needed to improve the results

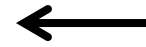
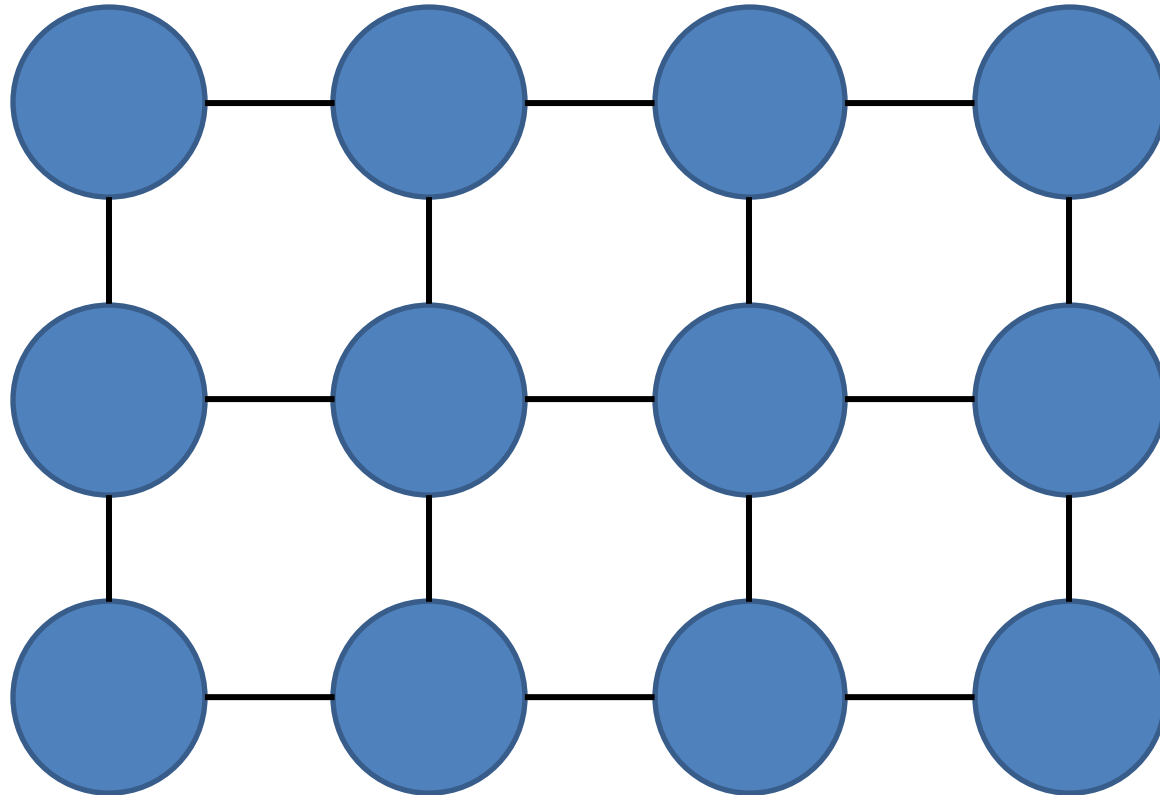


Basic segmentation algorithms

- These algorithms will not perform very well in complex images
- However, they can be useful for
 - **easy segmentation tasks** (e.g. controlled environments)
 - **initializing** or **autoseeding** more sophisticated algorithms

Additional methods (not presented in the class)

Image as a graph



Nodes: pixels



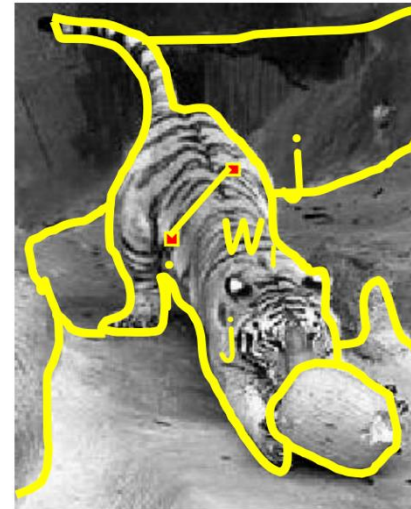
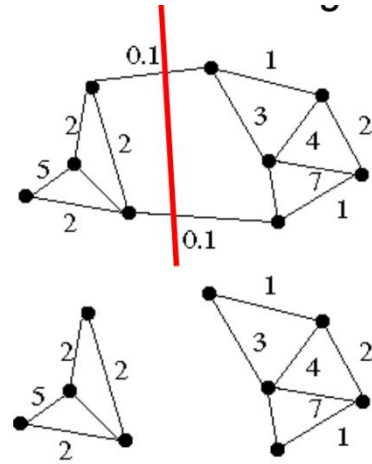
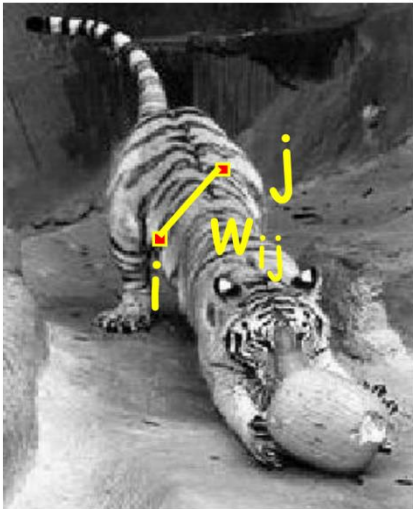
Edges: Constraints between
neighbouring pixels

Weights on edges reflect similarity
(affinity) in:

- Brightness
- Colour
- Texture
- Distance

Graph Cuts

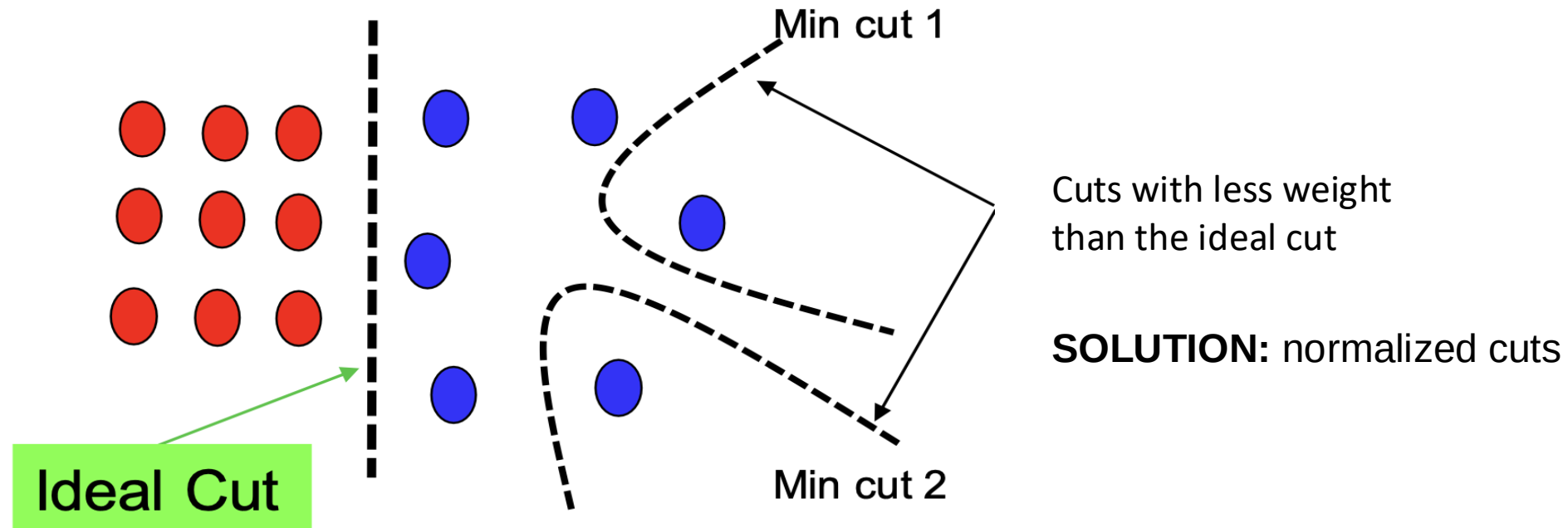
- Break graph into segments
 - Delete links that cross between segments
 - Easiest to **break links** that have **low affinity**
 - similar pixels should be in the same segments
 - dissimilar pixels should be in different segments



A cut of a graph G is the set of edges S such that removal of S from G disconnects G . **Minimum cut** is the cut of minimum weight

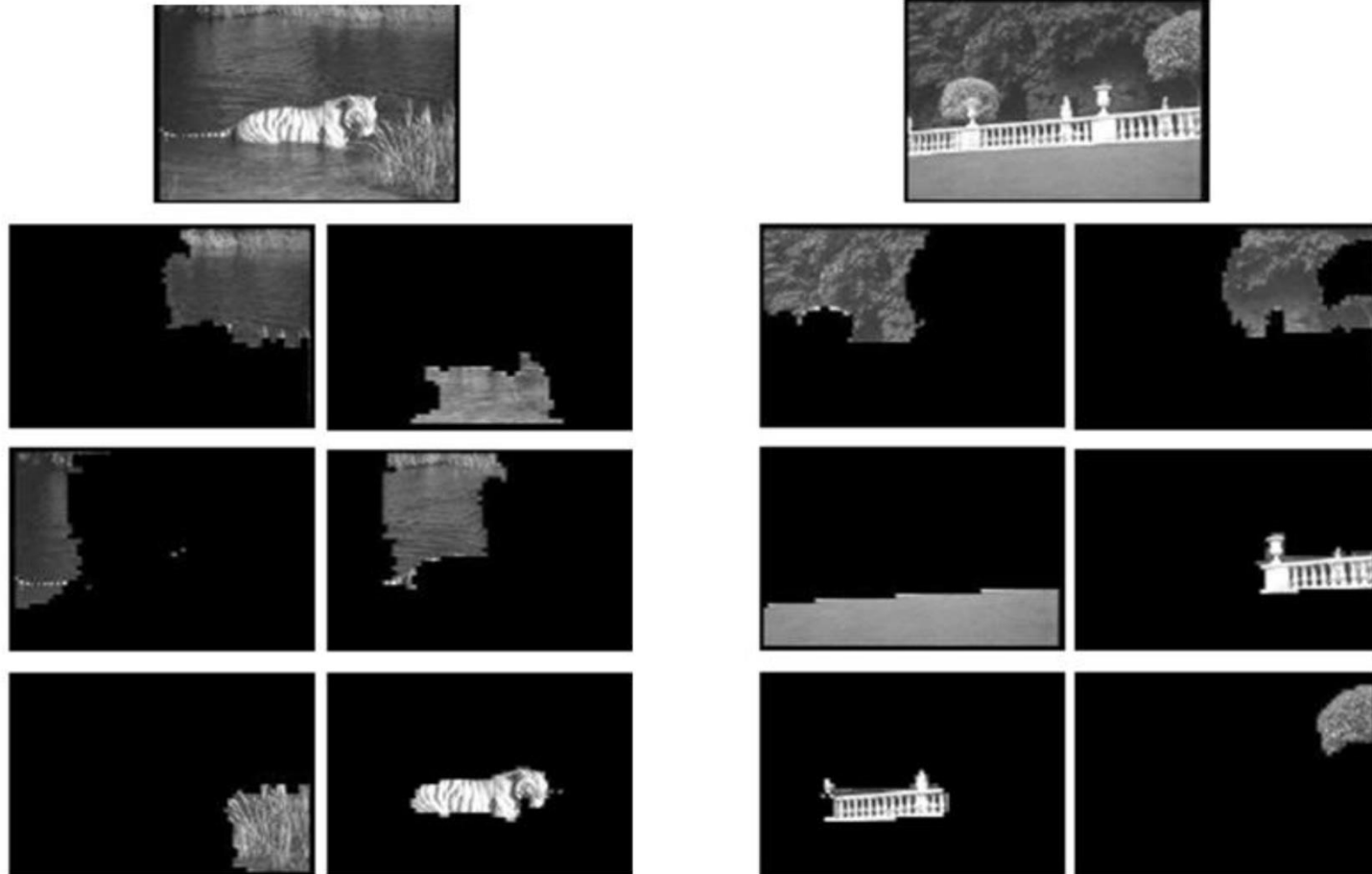
Graph Cuts

- There can be more than one minimum cut in a given graph.
- Using a **minimum cut** as a segmentation criterion, does not usually result in reasonable clusters, since the smallest cuts usually involve isolating a single pixel



Graph Cuts

- Example results with **normalized cuts**



GrabCut

GrabCut is a **mixture of two components**

1. Segmentation using graph cuts 
2. Foreground-background modeling using unsupervised clustering

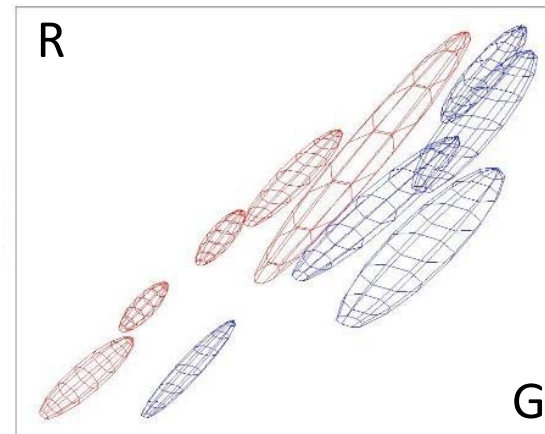
Foreground-background modeling

Given foreground/background labels



build a color model for both

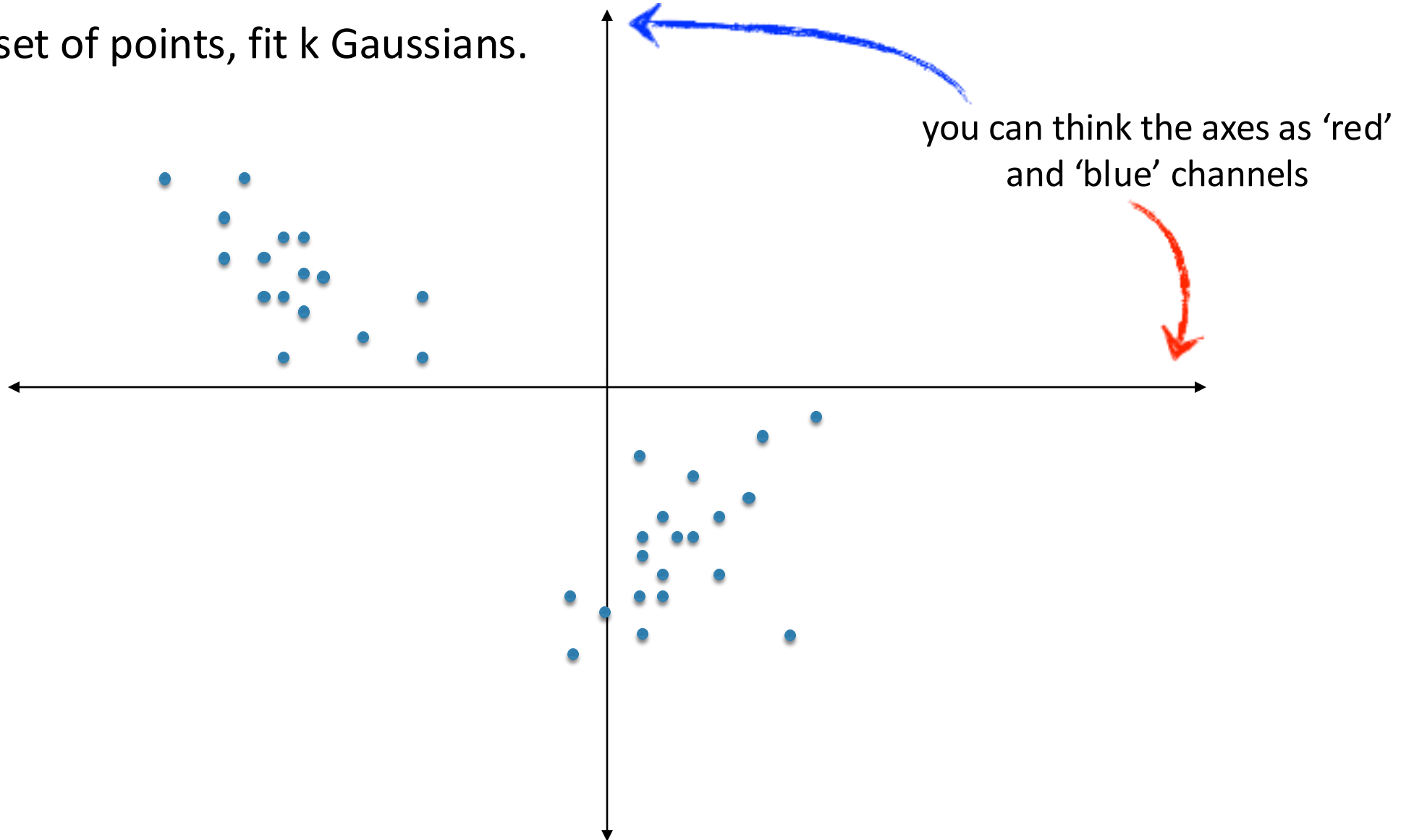
foreground



background

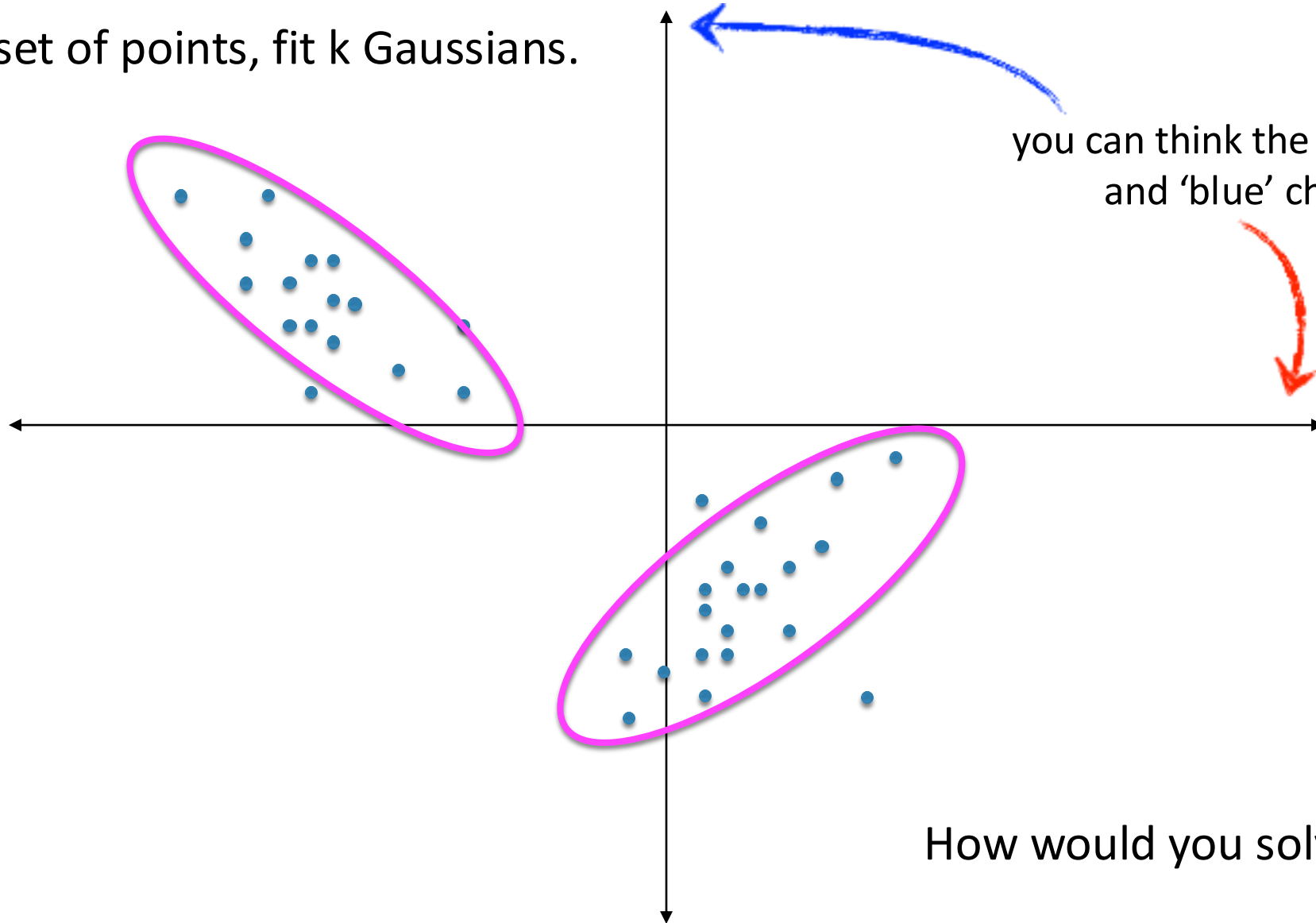
Learning color models

Given a set of points, fit k Gaussians.



Learning color models

Given a set of points, fit k Gaussians.



How would you solve this problem?

Expectation-Maximization: “soft” version of K-means

Given k :

1. Select initial centroids at random.

compute the probability of each object being in a cluster

E-step

2. Assign each object to the cluster with the nearest centroid.

and covariance

M-step

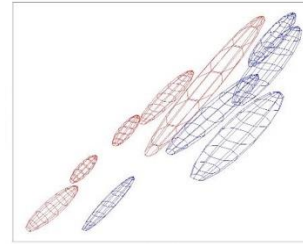
3. Compute each centroid as the mean of the objects assigned to it.

weighed by the probability of being in that cluster

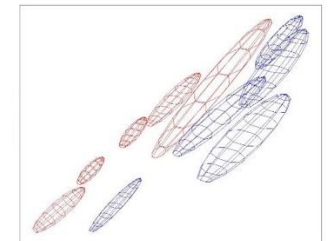
4. Repeat previous 2 steps until no change.

GrabCut

1. Segmentation using graph cuts
 - Requires having foreground model



2. Foreground-background modeling using unsupervised clustering
 - Requires having segmentation



What do we do?

iterate between two steps

GrabCut

Only user input is the box!



grab



cut paste

GrabCut - Iteration can be interactive

